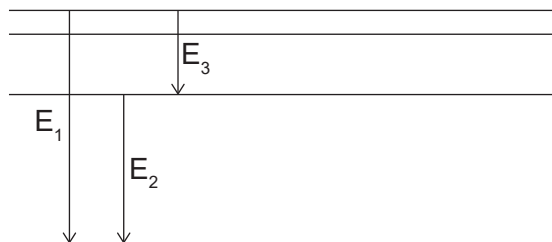


E.1 / PAPER 1 / QUESTION 14

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14. The diagram shows atomic transitions E_1 , E_2 and E_3 when a particular atom changes its energy state. The wavelengths of the photons that correspond to these transitions are λ_1 , λ_2 and λ_3 .



What is correct for these wavelengths?

- A. $\lambda_1 > \lambda_2 > \lambda_3$
- B. $\lambda_1 = \lambda_2 + \lambda_3$
- C. $\frac{1}{\lambda_1} = \frac{1}{\lambda_2 + \lambda_3}$
- D. $\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$